

Leva.Framework.Fakes

`Leva.Framework.Fakes` provides reusable test doubles and shared test helpers for Core, Engine, provider-neutral Identity, and provider-neutral Notifications libraries. It helps tests assemble deterministic runtime scenarios without duplicating production engine behavior.

Fakes exist for unit tests, integration-style framework tests, demos, and application tests that need predictable clocks, queues, events, logs, transitions, states, routines, behaviors, status updates, alarms, identity sources, authentication policies, authorization policies, notification gateway/storage behavior, result assertions, or a ready-made fake context. `Leva.Framework.Fakes` may depend on Core, Engine, Identity, and Notifications because it is a testing support library for those projects. Engine, Identity, and Notifications must never depend on Fakes. Fakes should not depend on application projects, UI providers, storage providers, notification providers, concrete authentication providers, or external infrastructure.

```
Leva.Framework.Fakes
  -> Leva.Framework.Core
  -> Leva.Framework.Engine
  -> Leva.Framework.Identity
  -> Leva.Framework.Notifications
```

Project overview

Fakes are intentionally small and observable. They expose what happened in a test: captured log entries, transition requests, queued events, current fake time, handler calls, enter/exit counts, status updates, alarm calls, identity/auth-session lookups, policy calls, notification sends, stored notification entries, configured return behavior, and failed `Result` assertions.

The source files are grouped by the framework area they support: `Core`, `Engine`, `Identity`, `Notifications`, and `Testing`. The namespace stays `Leva.Framework.Fakes` so tests can import one namespace and use all fakes and shared test helpers.

Use focused fakes when testing one component. For example, use `FakeClock` for time-dependent tests, `FakeTransition` for transition-request tests, `FakeLogSink` for log-output tests, `FakeEventQueue` for queue behavior, and `FakeNotificationGateway` for notification failure paths. Use `FakeContext` when a test needs convenient fake wiring and manual composition would distract from the test purpose.

Fakes should not become a second engine, second identity provider, or second notification provider. If a fake starts reproducing too much production behavior, the test should probably use the real library directly or introduce a smaller test seam.

Files and classes

Core fakes

`FakeAccess` - Simple access object for tests that exposes transition and log capabilities. `FakeClock` - Controllable clock for deterministic tests. `FakeEvent` - Simple event implementation with generated or supplied `EventId` and name. `FakeEventQueue` - Deterministic in-memory event queue implementation with priority ordering and async waiting. `FakeLogSink` - Stores log entries in memory for tests. `FakeTransition` - In-memory transition capability that captures requested transitions. `FakeTransitionEntry` - Represents one transition request captured by `FakeTransition`.

Engine fakes

`FakeAlarmSupervisor` - Configurable `IAAlarmSupervisor` fake with call tracking. `FakeBehavior` - Configurable behavior fake that records handling calls. `FakeContext` - Provides ready-to-use fake clock, log sink, transition, access, and event queue wiring for tests. `FakeRoutine` - Configurable routine fake with mutable lifecycle status. `FakeState` - Configurable state fake that records enter, exit, and handle calls. `FakeStatusUpdater` - Configurable `IStatusUpdater` fake with call tracking.

Identity fakes

`FakePrincipalStore` - In-memory principal store fake for tests. `FakeAuthSessionStore` - In-memory auth session store fake for tests. `FakeAuthSessionSource` - Configurable current auth session source fake for tests. `FakeAuthenticationPolicy` - Configurable authentication policy fake with call tracking. `FakeAuthorizationPolicy` - Configurable authorization policy fake with call tracking and simple policy matching.

Notification fakes

`FakeNotificationGateway` - Configurable notification gateway fake with captured sends and one-shot failure behavior. `FakeNotificationStore` - Configurable notification store fake with snapshots and one-shot save failure behavior.

Test helpers

`ResultAssert` - Provides shared assertions for successful `Result` and `Result<T>` values. `ResultAssertionException` - Exception thrown when a shared result assertion fails.